

Topic 41: Forces – Forces and motion

Students should:		Guidance <i>iLowerSecondary Curriculum reference</i>
41.1	understand the relationship between forces (balanced and unbalanced) on an object and its motion	understand the use of different-sized arrows to indicate the size and direction of action of a force explain why some objects float using forces of weight and upthrust explain the effects of balanced and unbalanced forces (This builds on <i>Year 7: Forces – Different types of forces</i> and <i>Year 8: Forces – More on types of forces</i>) <i>Year 9: Forces – Forces and motion</i>
41.2	understand the concept of a resultant force	<i>Year 9: Forces – Forces and motion</i>
41.3	understand how to perform simple calculations of resultant forces	<i>Year 9: Forces – Forces and motion</i>
41.4	understand the concept of terminal velocity of a falling object	for example, parachutist <i>Year 9: Forces – Forces and motion</i>
41.5	know that there are various units for speed and understand how to interconvert between them	for example, convert km/h into m/s <i>Year 9: Forces – Forces and motion</i>
41.6	understand how to use the formula: average speed = distance travelled ÷ time taken	understand the concepts of speed and average speed <i>Year 9: Forces – Forces and motion</i>
41.7	understand how to construct and interpret distance-time graphs, describing patterns or relationships	including evaluating gradient to calculate speed <i>Year 9: Forces – Forces and motion</i>